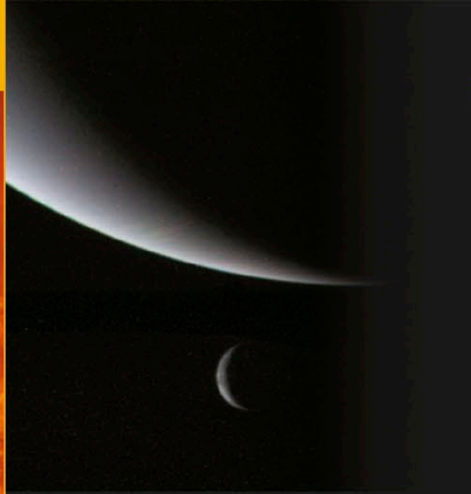
NEPTUNE'S MOONS



Neptune has only one major moon: Triton. All its other satellites are small and can be described as inner or outer moons depending on whether they are closer to or farther from Neptune than Triton. Six of the seven inner moons were discovered by analysis of Voyager 2 data in 1989. The moons are named after characters associated with the Roman god of the sea, Neptune, or his Greek counterpart, Poseidon.

NEPTUNE AND TRITON
This image of the crescent moon of Triton below the crescent of Neptune was captured by Voyager 2 as it flew away from the planet.

INNER MOON

Larissa

DISTANCE FROM NEPTUNE 45,617 miles (73,458 km)

ORBITAL PERIOD 0.55 Earth days

LENGTH 134 miles (216 km)

Larissa is the fifth moon from Neptune, lying outside the ring system. The moon was first spotted from Earth in 1981, but astronomers eventually decided that it was a ring arc circling Neptune. In late July 1989, a Voyager 2 team of astronomers confirmed that it is, in fact, an irregularly shaped, cratered moon. It was named after a lover of Poseidon.

IRREGULARLY SHAPED MOON

INNER MOON

Proteus

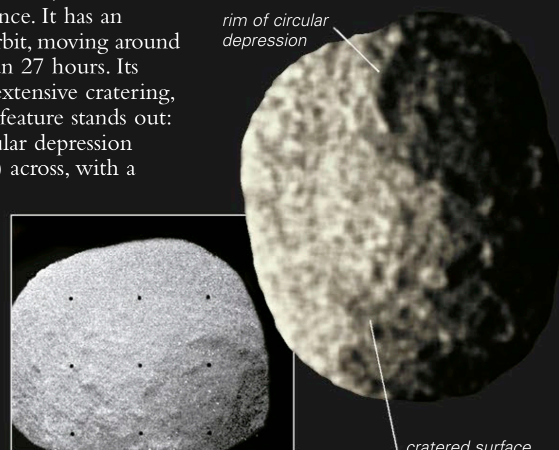
DISTANCE FROM NEPTUNE 73,059 miles (117,647 km)

ORBITAL PERIOD 1.12 Earth days

LENGTH 273 miles (440 km)

The most distant of the inner moons from Neptune, Proteus is also the largest of the seven—with the exception of S/2004 N1, their size increases with distance. It has an almost equatorial orbit, moving around Neptune in less than 27 hours. Its visible surface has extensive cratering, but just one major feature stands out: a large, almost circular depression 158 miles (255 km) across, with a rugged floor.

Proteus was the first of the inner moons to be found by Voyager 2 scientists. It was detected in mid-June 1989, within 2 months of the probe's closest approach to Neptune, enabling the observation sequence to be changed. The images later recorded by Voyager 2 revealed a gray, irregular, but roughly spheroid moon that reflects 6 percent of the sunlight hitting it. The moon was later named Proteus after a Greek sea god.



TWO VIEWS
The first image of Proteus (far right) shows the moon half-lit. The second was taken closer in. (The black dots are a processing artifact.)

OUTER MOON

Nereid

DISTANCE FROM NEPTUNE 3.4 million miles (5.5 million km)

ORBITAL PERIOD 360.1 Earth days

DIAMETER 211 miles (340 km)

Nereid was discovered on May 1, 1949, by the Dutch-born astronomer Gerard Kuiper while working at the McDonald Observatory in Texas. Little is still known about this moon. Voyager 2 flew by at a distance of 2.9 million miles (4.7 million km) in 1989 and could take only a low-resolution image. Nereid's outstanding characteristic is its highly eccentric and inclined orbit, which takes the moon out as far as about 5.9 million miles (9.5 million km) from Neptune and to within just 507,500 miles (817,200 km) at its closest approach.

BEST VIEW
Voyager 2 revealed Nereid to be a dark moon, reflecting only 14 percent of the sunlight it receives.

OUTER MOON

Halimede

DISTANCE FROM NEPTUNE 9.7 million miles (15.7 million km)

ORBITAL PERIOD 1,874.8 Earth days

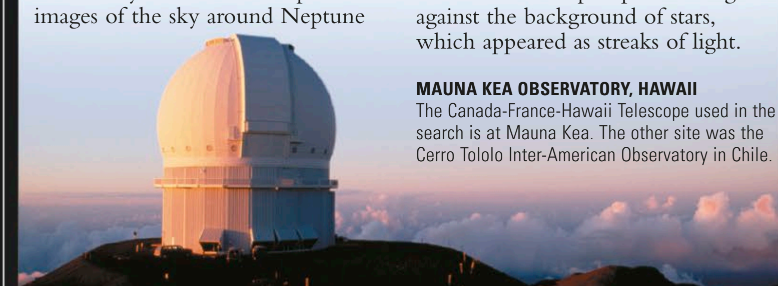
DIAMETER 30 miles (48 km)

Halimede was discovered by an international team of astronomers who were carrying out a systematic search for new Neptunian moons. Their task was not easy because moons as small and as distant as Halimede are extremely difficult to detect. Halimede follows a highly inclined and elliptical orbit. The origin of the irregular outer moons, which now number five, is unknown. More may be found, since these moons could be the result of an ancient collision between a former moon and a passing body such as a Kuiper Belt object.

EXPLORING SPACE

LOOKING FOR NEW MOONS

A team of astronomers announced the discovery of three new moons, including Halimede, on January 13, 2003. They had taken multiple images of the sky around Neptune from two sites in Hawaii and Chile. The images were combined to boost the signal of faint objects. The new moons showed up as points of light against the background of stars, which appeared as streaks of light.



MAUNA KEA OBSERVATORY, HAWAII
The Canada-France-Hawaii Telescope used in the search is at Mauna Kea. The other site was the Cerro Tololo Inter-American Observatory in Chile.

MAJOR MOON

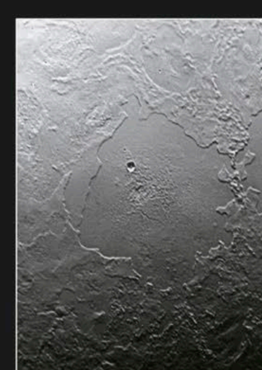
Triton

DISTANCE FROM NEPTUNE 220,306 miles (354,760 km)

ORBITAL PERIOD 5.88 Earth days

DIAMETER 1,681 miles (2,707 km)

Triton was the first of Neptune's moons to be discovered, just 17 days after the discovery of the planet was announced. William Lassell (see panel, right) used the coordinates published in *The Times* to locate Neptune in early October 1846. On October 10, he found its biggest moon using the 24-in (61-cm) reflecting telescope at his observatory in Liverpool, England. The moon was named Triton after the sea-god son of Poseidon. The Voyager 2 flyby nearly 143 years later revealed most of what is now known about this icy world.



SMOOTH PLAIN
The 185-mile (300-km) wide Ruach Planitia is in the cantaloupe terrain. It may be an old impact crater that has been filled in.